2024/2025

Anonymization of a Dataset with Utility and Risk Analysis

Security and Privacy

Authors	Francisca Cerqueira (202303764), Iara Ferreira (202305194), Rodrigo Simões (202303448)
Group	PL01_9

Table of Contents

1. Introduction 3
2. Attribute Classification 4
2.1 Attribute: sex 4
2.2 Attribute: age 4
2.3 Attribute: race 5
2.4 Attribute: marital-status 5
2.5 Attribute: education 5
2.6 Attribute: native-country 6
2.7 Attribute: workclass 6
2.8 Attribute: occupation 6
2.9 Attribute: salary-class 7
2.10 Summary of Attribute Classification 7
3. Risk Analysis of the Original Dataset 8
3.1 Attacker Models Overview 8
3.2 Risk Evaluation Results and Interpretation 8
3.3 Additional Risk Metrics (Original Dataset) 9
3.4 Population Uniqueness Estimation 10
4. Privacy Model Application and Evaluation 11
4.1 First Privacy Model Selected: k-Anonymity 11
4.2 Comparison of Re-identification Risk Before and After Anonymization Using k-Anonymity 11
4.3 Additional risk metrics after applying k-anonymity 12
4.4 Utility Evaluation – k-Anonymity 13
4.5 Impact of Varying *k* in k-Anonymity 14
..... 14
4.6 Second Privacy Model Selected: l-Diversity 17
4.7 Comparison of Re-identification Risk Before and After Anonymization Using Distinct l-Diversity 18
4.8 Additional risk metrics after applying distinct l-diversity 19
4.10 Utility Evaluation – Distinct l-Diversity 20
4.11 Impact of Varying *l* in Distinct l-Diversity 21
4.12 Comparison between k-Anonymity (k=5) and Distinct ℓ-Diversity(l=2) 22
5. Parameter Sensitivity Analysis of the Privacy Model 23
5.1 Suppression Limit 23
5.2 k-Anonymity (k = 5) – Suppression Limit 23
5.3 Distinct l-Diversity (l = 2) – Suppression Limit 26
5.4 - Effect of Coding Model on Privacy Risk and Utility 29
5.5 k-Anonymity (k = 5) - Effect of Coding Model on Privacy Risk and Utility 29
5.6 Distinct l-Diversity (l = 2) - Effect of Coding Model on Privacy Risk and Utility 30
5.7 Attribute Weights – Customizing the Importance of Quasi-Identifiers 31
5.8 k-Anonymity (k = 5) – Attribute Weights 31
5.9 – Distinct l-Diversity (l = 2) – Attribute Weights 34
6. Conclusion 35

1. Introduction

This assignment focuses on the anonymization of a real-world dataset using the ARX Data Anonymization Tool, with the goal of evaluating the trade-off between privacy protection and data utility. Following the methodology proposed in the lab instructions, we carried out a comprehensive analysis that includes the classification of attributes, assessment of re-identification risks, and the application of privacy models.

Initially, all attributes were classified into categories such as Identifying, Quasi-Identifying (QID), Sensitive, or Insensitive, based on their distinction and separation potential. Subsequently, the dataset's original privacy risks were evaluated using different attacker models available in ARX (Prosecutor, Journalist, and Marketer), which simulate adversarial attempts at re-identification under various assumptions.

We then applied two privacy models — k-anonymity and ℓ-diversity — to the dataset, comparing their performance in terms of re-identification risk reduction and data utility preservation. In addition, we conducted a parameter sensitivity analysis, exploring the impact of varying the suppression limit, the coding model and the attribute weights on both privacy and utility outcomes.

Based on our findings, we selected the privacy model configuration that offered the most effective balance between anonymity and analytical usefulness, which we submitted as our final ARX project.

2. Attribute Classification

In this section, each attribute of the dataset is analyzed and classified into one of the following categories:

- **Identifying (ID)**: attributes that directly identify individuals.
- **Quasi-Identifying (QID)**: attributes that, when combined, can lead to re-identification.
- **Sensitive (S)**: attributes that contain confidential or private information.
- **Insensitive (INS)**: attributes that are irrelevant for re-identification or analysis.

The classification considers the **distinction** and **separation** criteria, along with the attribute's semantic role and generalization hierarchy.

2.1 Attribute: sex

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: Male, Female
  Level-1: * (fully generalized)

The attribute sex is classified as a **Quasi-Identifying Attribute (QID)**. While it cannot identify an individual on its own, it can contribute significantly to re-identification when combined with other quasi-identifiers.

- **Distinction**: The attribute has only two distinct values, but it still contributes to narrowing down identity when linked with other attributes.
- **Separation**: Knowing someone's sex can increase the probability of distinguishing their record, especially in datasets where one sex is underrepresented.

2.2 Attribute: age

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: individual ages (e.g., 34)
  Level-1: ranges of 5 years (e.g., [31–35])
  Level-2: ranges of 10 years (e.g., [31–40])
  Level-3: ranges of 20 years (e.g., [21–40])
  Level-4: * (fully generalized)

The attribute age is classified as a **Quasi-Identifying Attribute (QID)** due to its potential to distinguish between individuals when combined with other attributes. It has high granularity and can significantly reduce anonymity if left unprotected. The generalization hierarchy defined for this attribute is robust and well-suited for anonymization. It starts with precise values and gradually groups ages into broader ranges.

- **Distinction**: Age has high cardinality (many distinct values), which increases its ability to distinguish between records.
- **Separation**: In specific populations (e.g., elderly, children), age can narrow down identity significantly.

2.3 Attribute: race

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: White, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, Other
  Level-1: * (fully generalized)

The race attribute is classified as a **Quasi-Identifying Attribute (QID)**. It cannot directly identify individuals, but it can reduce anonymity when combined with other QIDs. This is particularly relevant in datasets with imbalanced distributions across race categories.

Distinction: Race can have a significant impact on reducing the anonymity set, especially if certain categories are underrepresented.
Separation: It can increase the risk of re-identification when combined with other demographic variables, particularly in small or skewed populations.

2.4 Attribute: marital-status

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: specific values (Married-civ-spouse, Divorced, Separated, Never-married, Widowed, Married-spouse-absent, Married-AF-spouse)
  Level-1: Spouse present / Spouse not present
  Level-2: * (fully generalized)

The marital-status attribute is classified as a **Quasi-Identifying Attribute (QID)** due to its indirect role in re-identification when combined with other demographic variables. Its generalization strategy reflects the presence or absence of a spouse, which allows for a balance between privacy and analytical value.

- **Distinction**: Marital status helps reduce the anonymity set, especially in populations where certain statuses (e.g., widowed) are rare.
- **Separation**: It can help isolate individuals based on demographic patterns, contributing to re-identification risk when used in conjunction with other QIDs.

2.5 Attribute: education

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: specific degrees (e.g., HS-grad, Doctorate)
  Level-1: educational stages (e.g., High School, Undergraduate)
  Level-2: grouped categories (Primary education, Secondary education, Higher education)
  Level-3: * (fully generalized)

The education attribute is classified as a **Quasi-Identifying Attribute (QID)**. While not uniquely identifying, it provides high distinction and contributes meaningfully to re-identification risk when combined with other variables. The hierarchical structure supports flexible generalization, which helps protect privacy while retaining analytic value.

- **Distinction**: Education level often correlates with age, income, and other personal characteristics. Some education levels are uncommon and may isolate individuals.
- **Separation**: Education level may indirectly help infer other private details (like profession or socioeconomic status), increasing re-identification probability.

2.6 Attribute: native-country

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: country-level (e.g., India, Germany, Cuba)
  Level-1: regional grouping (Asia, Europe, North America, South America, Africa)
  Level-2: * (fully generalized)

The native-country attribute is classified as a **Quasi-Identifying Attribute (QID)**. While it does not identify an individual directly, it can significantly reduce anonymity when combined with other quasi-identifiers, particularly in the case of underrepresented countries or regions. The generalization strategy used — grouping countries into regions — is appropriate for anonymization and helps reduce identification risk while preserving analytical relevance.

- **Distinction**: Country of origin is often highly distinguishing, especially for countries with small representation in the dataset.
- **Separation**: The attribute can help infer ethnic, cultural, or linguistic characteristics that may further assist in re-identification.

2.7 Attribute: workclass

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: specific employment types (e.g., Private, Federal-gov, Never-worked)
  Level-1: broader groups (Government, Non-Government, Unemployed)
  Level-2: * (fully generalized)

The workclass attribute is classified as a **Quasi-Identifying Attribute (QID)**. While it does not uniquely identify individuals, it may contribute to re-identification when combined with other QIDs. For instance, certain employment types are less common and can act as distinguishing factors. The hierarchy defined supports privacy-preserving transformations by grouping work categories into higher-level abstractions while retaining analytical potential.

- **Distinction**: Certain workclass categories are relatively rare or specific (e.g., Without-pay, Federal-gov), which can increase identifiability.
- **Separation**: It can indirectly suggest socioeconomic status or professional sector, adding to the risk of re-identification.

2.8 Attribute: occupation

- **Type**: Quasi-Identifying Attribute
- **Generalization hierarchy**:
  Level-0: specific job roles (e.g., Sales, Tech-support, Farming-fishing)
  Level-1: categorized as Technical, Nontechnical, or Other
  Level-2: * (fully generalized)

The occupation attribute is classified as a **Quasi-Identifying Attribute (QID)**. Although it does not directly identify individuals, it provides relevant distinguishing information when combined with other quasi-identifiers. Certain roles may be uncommon in the dataset, reducing the anonymity set. The generalization scheme allows for flexible anonymization by grouping occupations into broader sectors, making it suitable for privacy-preserving analysis without entirely sacrificing utility.

- **Distinction**: Some occupations are uncommon or specific (e.g., Armed-Forces, Priv-house-serv) and thus highly identifying in certain contexts.
- **Separation**: Occupation can also suggest income level, industry, or education background, increasing the re-identification risk in combination with other QIDs.

2.9 Attribute: salary-class

- **Type:** Dual role – Quasi-Identifying Attribute (QID) for *k-anonymity*, Sensitive Attribute (SA) for *l-diversity*
- **Generalization hierarchy**:
  Level-0: ≤50K, >50K
  Level-1: * (fully generalized)

- **Dual Role Justification:**

  The attribute salary-class reflects income information and is therefore typically considered sensitive. In this assignment, it serves a dual purpose based on the anonymization model applied:

  - Under the **k-anonymity model**, salary-class is treated as a **Quasi-Identifying Attribute (QID).** Although it has already been generalized into two broad income brackets, it remains a strong discriminator when combined with other quasi-identifiers. This allows the analysis of identity disclosure risks and utility trade-offs, leveraging the attribute's distinguishing capacity without requiring additional sensitivity constraints.
  - Under **the l-diversity model**, salary-class is treated as a **Sensitive Attribute (SA**). Even though its values are generalized, income level can still reflect personal socioeconomic information and correlate with characteristics such as education level, job type, or nationality. This justifies its classification as sensitive when analyzing diversity within equivalence classes to prevent inference attacks.

- **Distinction:**
  Despite containing only two categories, salary-class contributes meaningfully to the uniqueness of records. When used as a QID, it enhances profile differentiation, thereby influencing the granularity of anonymity sets.

- **Separation**:
  As an SA, salary-class may still enable indirect inferences about individuals' broader personal characteristics. Thus, ensuring adequate diversity in its values across equivalence classes becomes critical under the l-diversity model.

2.10 Summary of Attribute Classification

In conclusion, the attributes of the dataset were classified according to their potential contribution to re-identification, based on the principles of distinction and separation. Most variables were identified as **Quasi-Identifying Attributes (QIDs)**. The attribute salary-class, however, plays a **dual role**: it is considered a QID under the **k-anonymity** model and a **Sensitive Attribute** under the **l-diversity** model. This classification lays the foundation for the next step, in which the dataset's original privacy risks will be evaluated prior to the application of any anonymization models

3. Risk Analysis of the Original Dataset

3.1 Attacker Models Overview

The ARX tool provides risk analysis based on different attacker models, which simulate how an external adversary might attempt to re-identify individuals in the dataset. Each model reflects a different assumption about the attacker's knowledge and intent:

- **Prosecutor model:** Assumes the attacker knows that a specific person is in the dataset and attempts to re-identify their record.
- **Journalist model:** Assumes the attacker does not know whether a particular individual is in the dataset but still attempts to find matches.
- **Marketer model:** Simulates mass re-identification attempts based on probabilistic guesses, focusing on broad patterns rather than specific targets.

These models help quantify the re-identification risk under various threat scenarios, providing a comprehensive understanding of the dataset's vulnerability.

3.2 Risk Evaluation Results and Interpretation

The results obtained from ARX's attacker models show a high level of re-identification risk in the dataset before anonymization. The three models consistently point to significant vulnerabilities:

- **Prosecutor model**:
  - **Records at risk**: 93.20%
  - **Highest risk**: 100%
  - **Success rate**: 80.34%

This model reveals that nearly **93% of the records are vulnerable to direct re-identification**, assuming an attacker knows that the target is in the dataset. The fact that **100% is reached as the highest risk** confirms that some individuals are uniquely identifiable. The success rate of 80,34% indicates a high probability of successful attacks.

- **Journalist model**:
  - **Records at risk**: 93.29%
  - **Highest risk**: 100%
  - **Success rate**: 80.34%

Similar to the prosecutor model, the journalist model also detects significant risk, even when the attacker is unsure whether a person is included in the dataset. This suggests that anonymity is not preserved in many cases, even under weaker attacker assumptions.

- **Marketer model**:
  - **Success rate**: 80.34%

The marketer model, which focuses on broad probabilistic matching, achieves a success rate above 80%. This demonstrates that the dataset is highly susceptible to inference-based mass re-identification attacks.

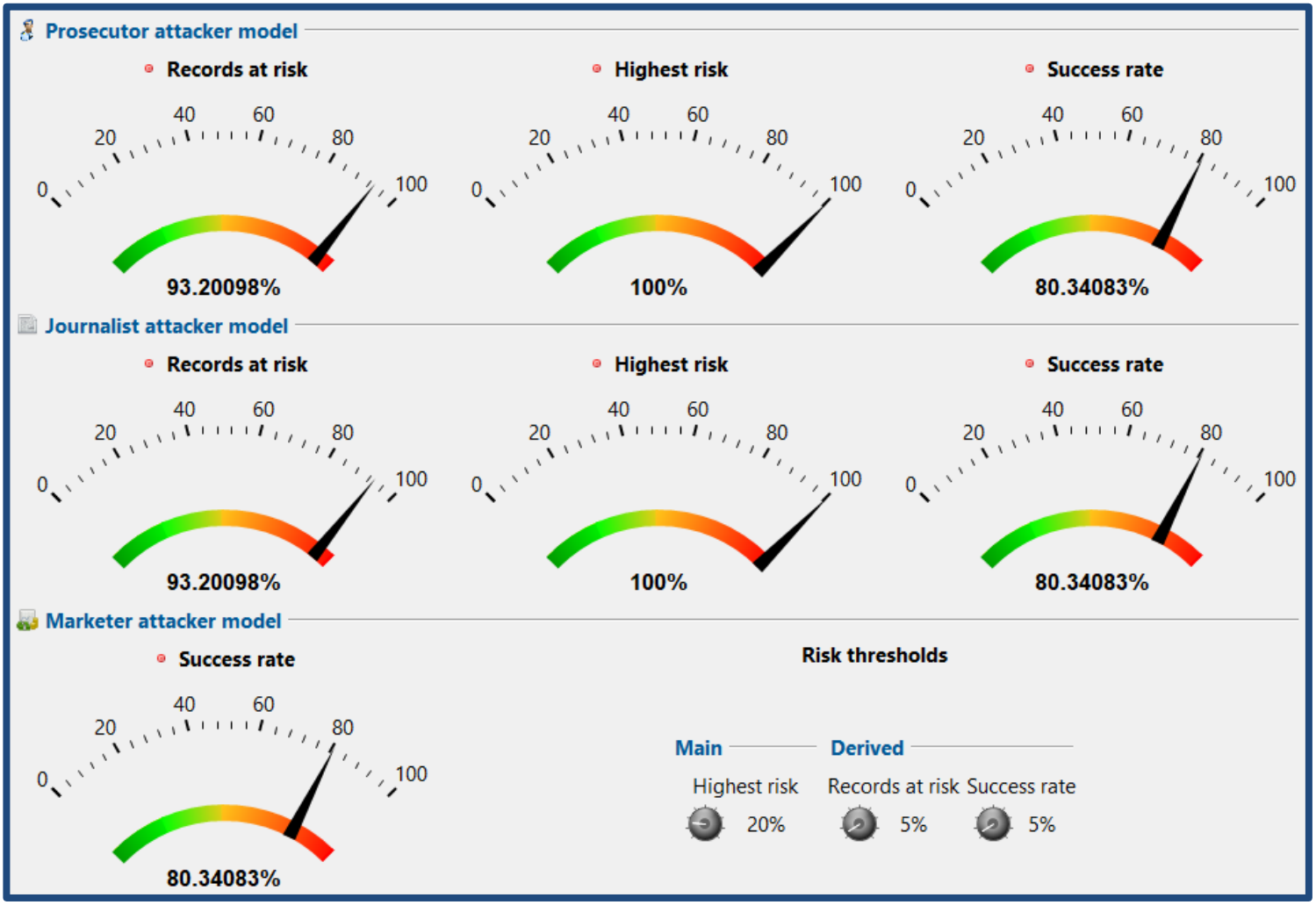

Figure 1 shows the graphical representation of the attacker model results. The visual indicators confirm a critical privacy risk in the dataset's original state.

3.3 Additional Risk Metrics (Original Dataset)

Metric	Value	Explanation
Lowest prosecutor risk	5.56	Minimum re-identification risk observed for any record
Records affected by lowest risk	0.32	Only a small fraction of records had the lowest possible risk
Average prosecutor risk	80.34%	Likelihood of correct re-identification across all records
Records affected by highest risk	69.33%	Portion of records with 100% re-identification risk
Estimated prosecutor risk	100%	Maximum likelihood of attacker success in targeted re-ID
Estimated journalist risk	100%	Risk under assumption of uncertain presence of the target
Estimated marketer risk	80.34%	Risk of probabilistic mass re-identification
Sample uniques	69.33%	Records that are unique within the dataset sample
Population uniques	4.17%	Records estimated to be unique in the general population
Population model used	PITMAN	Statistical model used to estimate population uniqueness

Table 1 Additional Risk Metrics

Additional extended metrics reported by ARX are summarized in Table 1. These indicators reinforce the conclusion that the dataset in its original form presents a high privacy risk and justifies the need for anonymization.

3.4 Population Uniqueness Estimation

In addition to the re-identification risks measured through attacker models, ARX also estimates the percentage of individuals likely to be unique in the broader population. This estimation is based on statistical models applied to varying sampling fractions.

Figure 2 illustrates the population uniqueness curve using the **Pitman**, **Dankar**, **Zayatz**, and **SNB** models. The **Zayatz** model estimates a constant uniqueness level around **50%**, while the **Pitman** and **Dankar** models show a steady increase in uniqueness as the sampling fraction grows. By contrast, the **SNB** model reports near-zero uniqueness, except for very large samples, making it less reliable in this context.

These results indicate that a significant proportion of individuals in the dataset are likely to be **unique in the general population**, not just within the sample. This reinforces the conclusion that the dataset, in its original state, presents a substantial re-identification risk — even when assessed using conservative population models.

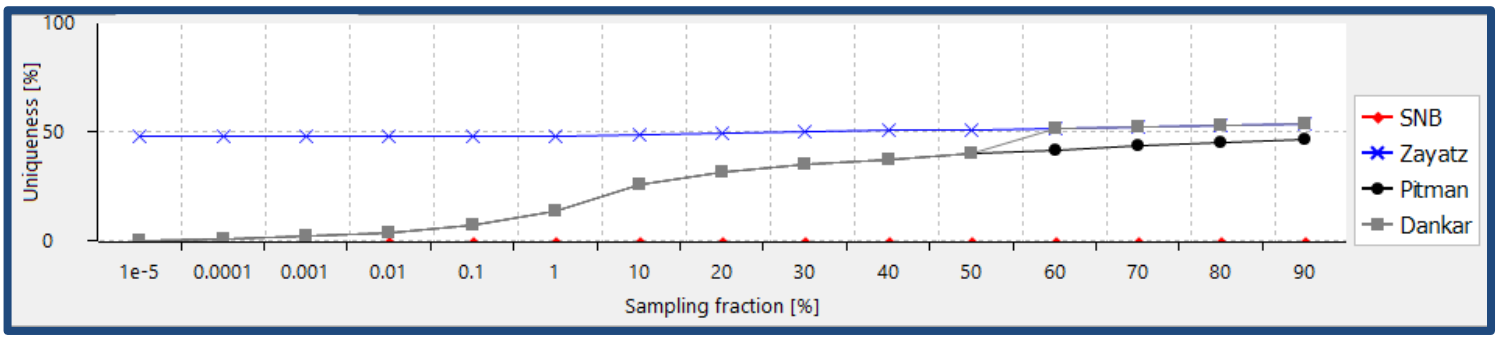

Figure 2 – Estimated population uniqueness by sampling fraction, based on multiple statistical models.

4. Privacy Model Application and Evaluation

4.1 First Privacy Model Selected: k-Anonymity

For this assignment, the first privacy model we chose was **k-anonymity**, a foundational model in privacy-preserving data publishing. The goal of k-anonymity is to ensure that each individual record in the dataset is **indistinguishable from at least (k–1) other records** with respect to a defined set of **quasi-identifying attributes (QIDs)**.

In this model, **all attributes were classified as quasi-identifiers (QIDs)**, including demographic, occupational, and socioeconomic variables. Since no attributes are treated as sensitive(in this case), there is no need to apply models that specifically protect sensitive information at the same time we apply this model.
This makes **k-anonymity the most appropriate and effective choice** for achieving privacy protection without unnecessary complexity.

The k-anonymity model was selected for the following reasons:

- **Complete applicability to the dataset**: With all attributes acting as QIDs, the k-anonymity model aligns naturally with the data structure, offering a straightforward yet powerful method of generalization and suppression to prevent identity disclosure.
- **Controlled and explainable privacy guarantee**: The model provides a clear privacy guarantee — no record is unique among fewer than *k* individuals. This makes it easy to understand and to justify in risk analysis, both for academic evaluation and practical deployment.
- **Balanced trade-off between privacy and utility**: K-anonymity allows flexible control of data generalization, enabling the minimization of information loss while satisfying privacy constraints. It is particularly effective for use cases where data utility must be preserved for analysis.

In this evaluation, the model was applied using a fixed value of **k = 5**, which represents a commonly accepted threshold for meaningful anonymization. The impact of this transformation was assessed through a comparative analysis of **privacy risk** and **data utility**, as detailed in the next sections.

The anonymization process was executed using ARX's **Optimal search strategy**, which guarantees the selection of a transformation that minimizes information loss while satisfying the privacy model. The **Global Transformation** model was used, applying a uniform generalization level across all records. This configuration ensures consistent treatment of attributes and promotes stronger privacy guarantees. These settings were chosen to support a high-quality anonymization process suitable for reliable evaluation in terms of both **re-identification risk** and **data utility**.

4.2 Comparison of Re-identification Risk Before and After Anonymization Using k-Anonymity

After applying k-anonymity with ***k = 5,*** the dataset was re-evaluated to measure its new privacy risk and compare it to the original (non-anonymized) version. The results indicate a **significant improvement in privacy protection**.

The **Prosecutor and Journalist attacker models** show that:

- **Records at risk** dropped from **93.20%** to **0%**,
- **Highest risk** decreased from **100%** to **20%**,
- **Success rate** fell from **80.34%** to **3.52%**.

Likewise, the **Marketer model** saw a dramatic drop in its success rate, from **80.34%** to **3.52%**.

These results clearly demonstrate that the anonymization was **highly effective** in reducing re-identification risks across all attacker types. Not only was the risk minimized, but **no individual records remained uniquely identifiable** within the dataset, satisfying the goal of k-anonymity.

Despite this, it's important to recognize that the transformation likely introduced some loss in data utility. This trade-off between privacy and data quality is expected and will be formally evaluated through utility metrics in the following subsection.
Overall, the application of k-anonymity with *k = 5* provided a strong privacy guarantee with a low success rate for potential attacks, aligning with the goals of the anonymization process.

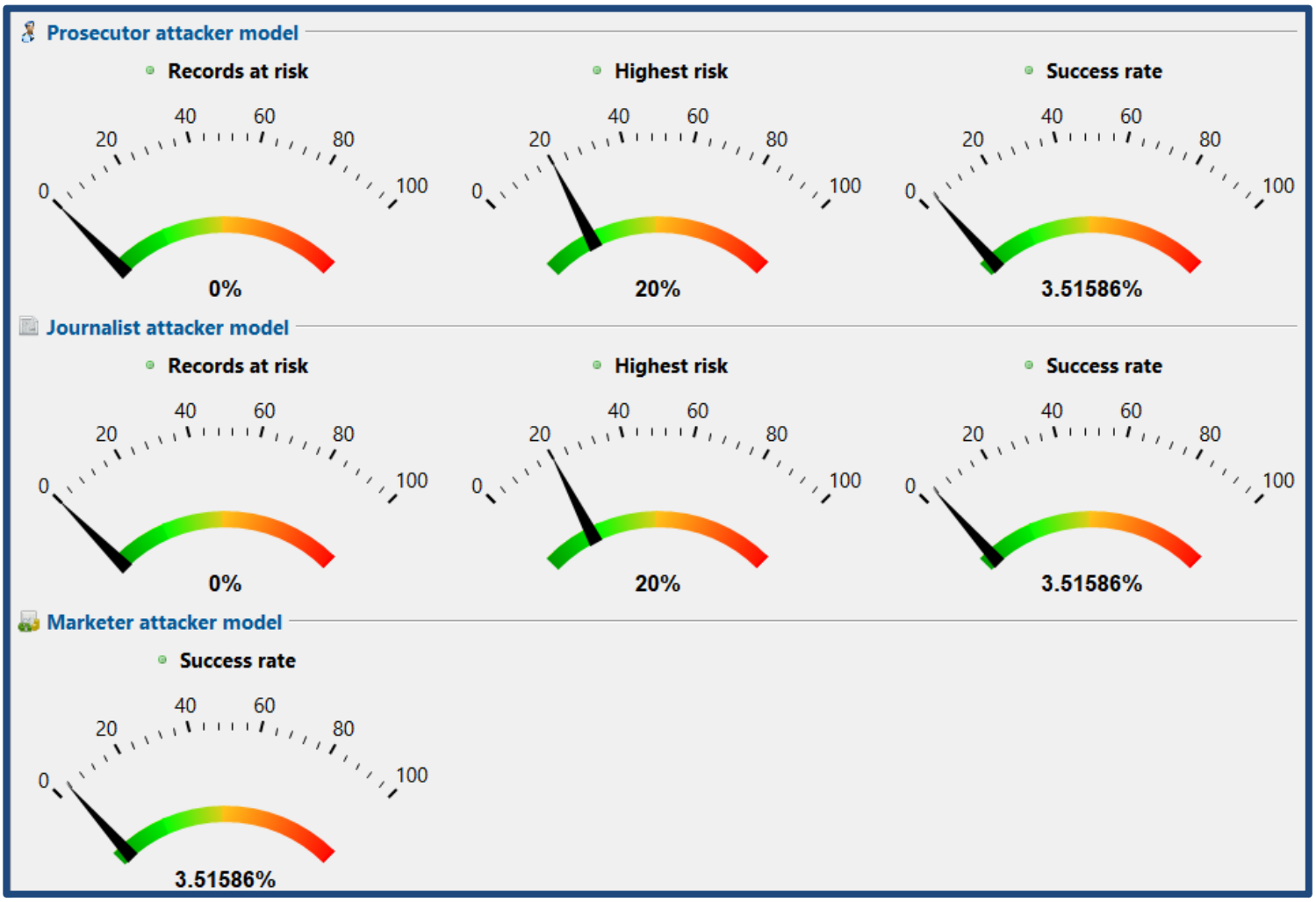

Figure 3 – Re-identification risk after applying k-anonymity with k = 5.

4.3 Additional risk metrics after applying k-anonymity

Metric	Value	Interpretation
Lowest prosecutor risk	0.32%	Minimum risk across all records
Records affected by lowest risk	6.28%	Small subset of records with near-zero risk
Average prosecutor risk	3.52%	Overall risk across all records
Highest prosecutor risk	20%	Maximum risk allowed under k-anonymity
Records affected by highest risk	2.22%	Only a small fraction reach this limit
Estimated prosecutor risk	20%	Consistent with max allowed
Estimated journalist risk	20%	Same as above
Estimated marketer risk	3.52%	Probabilistic attacker performance
Sample uniques	0%	No uniquely identifiable records remain
Population uniques	0%	Estimated risk in the general population
Population model	DANKAR	Statistical model used for uniqueness estimation

Table 2 - Additional Risk Metrics after anonymization with k-anonymity

Additional metrics provided by ARX reinforce the effectiveness of the anonymization. The average prosecutor risk dropped to 3.52%, while the maximum risk was capped at 20%, affecting only 2.22% of the records. Notably, no sample or population uniques remain, meaning all records now belong to indistinguishable anonymity groups. These results confirm that k-anonymity with *k = 5* effectively eliminates high-risk outliers while preserving overall data utility. The risk estimations were calculated using the DANKAR population model.

4.4 Utility Evaluation – k-Anonymity

Dataset-level quality

Model	Quality
Gen. intensity	56.35427%
Granularity	69.08781%
N.-U. entropy	38.8754%
Discernibility	85.58982%
Average class size	99.52207%
Record-level squared error	32.79373%
Attribute-level squared error	42.43219%
Aggregation-specific squared error	0%

Figure 4 - Dataset-level quality

Following the application of k-anonymity with *k = 5*, the utility of the anonymized dataset was assessed using several standard quality models provided by ARX. The results reflect a **moderate trade-off between privacy and data utility**, with some generalization necessary to achieve full compliance with the privacy model.

At the **dataset level**, the **generalization intensity** was measured at **56.35%,** indicating that just over half of the attribute values were generalized to meet the privacy requirement. The **granularity score** was **69.09%,** reflecting a reasonable preservation of data specificity across attributes.

The **non-uniform entropy** metric — which captures information loss in terms of value distributions — was **38.88%,** suggesting that although information has been suppressed or generalized, a considerable portion of the original distributional patterns is retained.

The **discernibility score** was relatively high at **85.59%,** meaning that many records remain distinguishable within their anonymity groups, despite being generalized. Additionally, the **average class size** was **99.52%,** showing that most equivalence classes formed during generalization are relatively small and tight, which typically favors utility.

Other utility indicators, such as the **record-level squared error** (32.79%) and **attribute-level squared error** (42.43%), also fall within expected ranges for this level of anonymization.

Overall, the results show that k-anonymity with *k = 5* achieved strong privacy protection while maintaining **reasonable data utility**, making the anonymized dataset viable for analysis tasks that tolerate moderate generalization.

Attribute-level quality

Attribute	Data type	Missings	Gen. intensity	Granularity	N.-U. entropy	Squared error
sex	String	13.05341%	86.94659%	86.94659%	85.67296%	86.94659%
age	String	13.05341%	43.4733%	79.04236%	21.67824%	83.77648%
race	String	13.05341%	86.94659%	86.94659%	66.20484%	88.40466%
marital-status	String	13.05341%	43.4733%	56.70532%	49.30235%	3.73048%
education	String	13.05341%	28.9822%	55.28578%	31.46791%	28.25649%
native-country	String	13.05341%	43.4733%	63.05341%	20.6638%	13.03487%
workclass	String	13.05341%	43.4733%	62.10471%	30.47818%	78.22999%
occupation	String	13.05341%	43.4733%	62.1412%	40.95812%	48.19992%
salary-class	String	13.05341%	86.94659%	86.94659%	87.17216%	86.94659%

Figure 5 - Attribute-level quality

At the **attribute level**, the impact of anonymization varied depending on the attribute's structure and generalization hierarchy. Variables such as sex, race, and salary-class exhibit **high generalization intensity and squared error** (≈86.95%), due to their limited value range and shallow hierarchies. These attributes were more heavily transformed, which reduces their precision. In contrast, attributes like education and native-country had **lower generalization and error scores**, preserving more detailed information. The attribute marital-status shows particularly low squared error (**3.73%**), suggesting that its granularity was well retained.

Overall, this confirms that **anonymization did not uniformly affect all variables**, and that **core demographic and categorical features remain analyzable despite generalization**.

4.5 Impact of Varying *k* in k-Anonymity

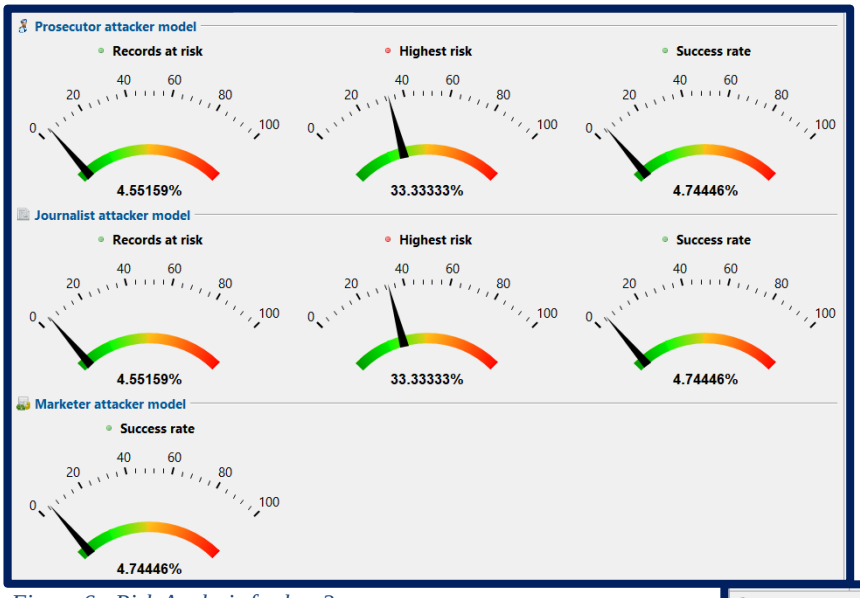

Figure 6 - Risk Analysis for k = 3

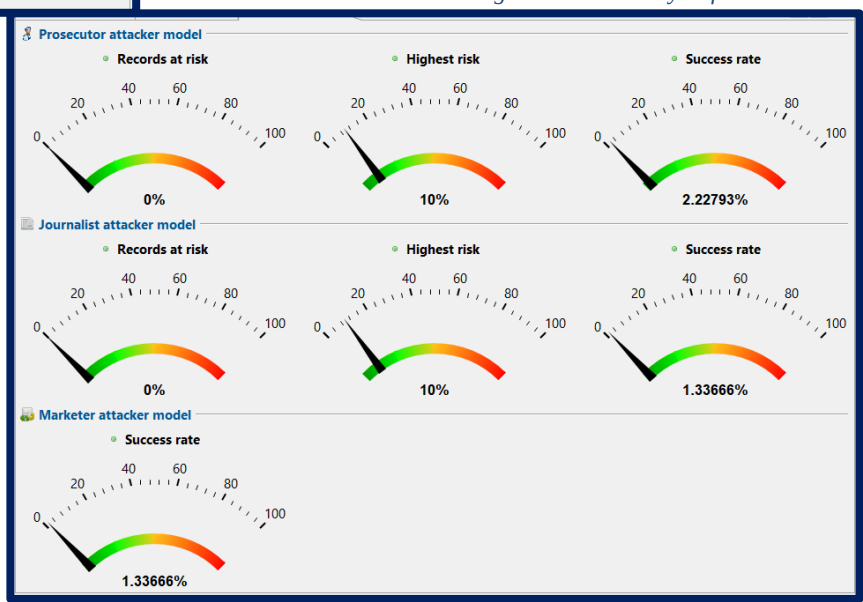

Figure 4 - Risk Analysis for k = 10

Attribute-level quality

Attribute	Data type	Missings	Gen. intensity	Granularity	N.-U. entropy	Squared error
sex	String	17.99016%	82.00984%	82.00984%	79.67646%	82.00984%
age	String	17.99016%	20.50246%	66.27058%	16.9486%	65.92245%
race	String	17.99016%	82.00984%	82.00984%	54.88071%	83.48106%
marital-status	String	17.99016%	41.00492%	53.58398%	46.40299%	3.53449%
education	String	17.99016%	27.33661%	52.09534%	29.37544%	26.49229%
native-country	String	17.99016%	41.00492%	59.46152%	15.96862%	12.36712%
workclass	String	17.99016%	41.00492%	58.57846%	27.84859%	74.30205%
occupation	String	17.99016%	41.00492%	58.53965%	38.61011%	45.73844%
salary-class	String	17.99016%	82.00984%	82.00984%	82.08142%	82.00984%

Dataset-level quality

Model	Quality
Gen. intensity	50.87647%
Granularity	64.04549%
N.-U. entropy	35.11918%
Discernibility	80.57399%
Average class size	99.23314%
Record-level squared error	29.96794%
Attribute-level squared error	35.16309%
Aggregation-specific squared error	0%

Figure 8 - Utility Metrics for k = 10

Attribute-level quality

Attribute	Data type	Missings	Gen. intensity	Granularity	N.-U. entropy	Squared error
sex	String	8.90724%	91.09276%	91.09276%	90.25965%	91.09276%
age	String	8.90724%	45.54638%	82.8116%	23.2749%	87.35585%
race	String	8.90724%	91.09276%	91.09276%	75.25257%	92.38937%
marital-status	String	8.90724%	45.54638%	59.10635%	51.78946%	4.03823%
education	String	8.90724%	30.36425%	58.05458%	33.45796%	30.03911%
native-country	String	8.90724%	45.54638%	66.07344%	25.51216%	13.56488%
workclass	String	8.90724%	45.54638%	65.06626%	32.6693%	81.53183%
occupation	String	8.90724%	45.54638%	65.08054%	42.9161%	50.30892%
salary-class	String	8.90724%	91.09276%	91.09276%	91.14382%	91.09276%

Dataset-level quality

Model	Quality
Gen. intensity	59.0416%
Granularity	72.36268%
N.-U. entropy	41.45531%
Discernibility	89.73487%
Average class size	99.6515%
Record-level squared error	34.48329%
Attribute-level squared error	44.26599%
Aggregation-specific squared error	0%

Figure 9 - Utility Metrics for k = 3

To evaluate how the value of *k* in k-anonymity impacts privacy and utility, the model was applied using *k = 3*, *k = 5*, and *k = 10*. As expected, increasing *k* led to a consistent **reduction in re-identification risk** across all attacker models. At *k = 3*, the highest risk remained at 33.33%, and 4.55% of records were still at risk. By contrast, *k = 10* reduced the highest risk to just 10%, with **0% of records at risk** and success rates as low as **1.34%** for the journalist and marketer models.

On the utility side, increasing *k* led to **greater generalization** and slightly more information loss. Generalization intensity decreased from **59.04%** at *k = 3* to **50.88%** at *k = 10*, while non-uniform entropy declined moderately from **41.46%** to **35.12%**. Discernibility also dropped, indicating that fewer records have retained detailed value combinations. Still, all utility scores remained within acceptable ranges, and the dataset preserved significant analytical value even at higher privacy levels.

Overall, these results highlight the **privacy–utility trade-off** inherent in k-anonymity: while higher *k* values yield stronger protection, they come at the cost of reduced data utility. Selecting the appropriate *k* depends on the balance between minimizing disclosure risk and maintaining analytical usefulness for downstream tasks. So k=5 is a great balance between data protection and data utility.

k	Records at Risk	Highest Risk	Success Rate
3	4.55%	33.33%	4.74%
5	0%	20%	3.52%
10	0%	10%	2.23% (Prosecutor), 1.34% (Journalist/Marketer)

Table 3 - Risk Metrics by k Value

k	Gen. Intensity	Granularity	N.-U. Entropy	Discernibility	Record SE	Attr. SE
3	59.04%	72.36%	41.46%	89.73%	34.48%	44.27%
5	56.35%	69.09%	38.88%	85.59%	32.79%	42.43%
10	50.88%	64.05%	35.12%	80.57%	29.97%	35.16%

Table 4 - Utility Metrics by k Value

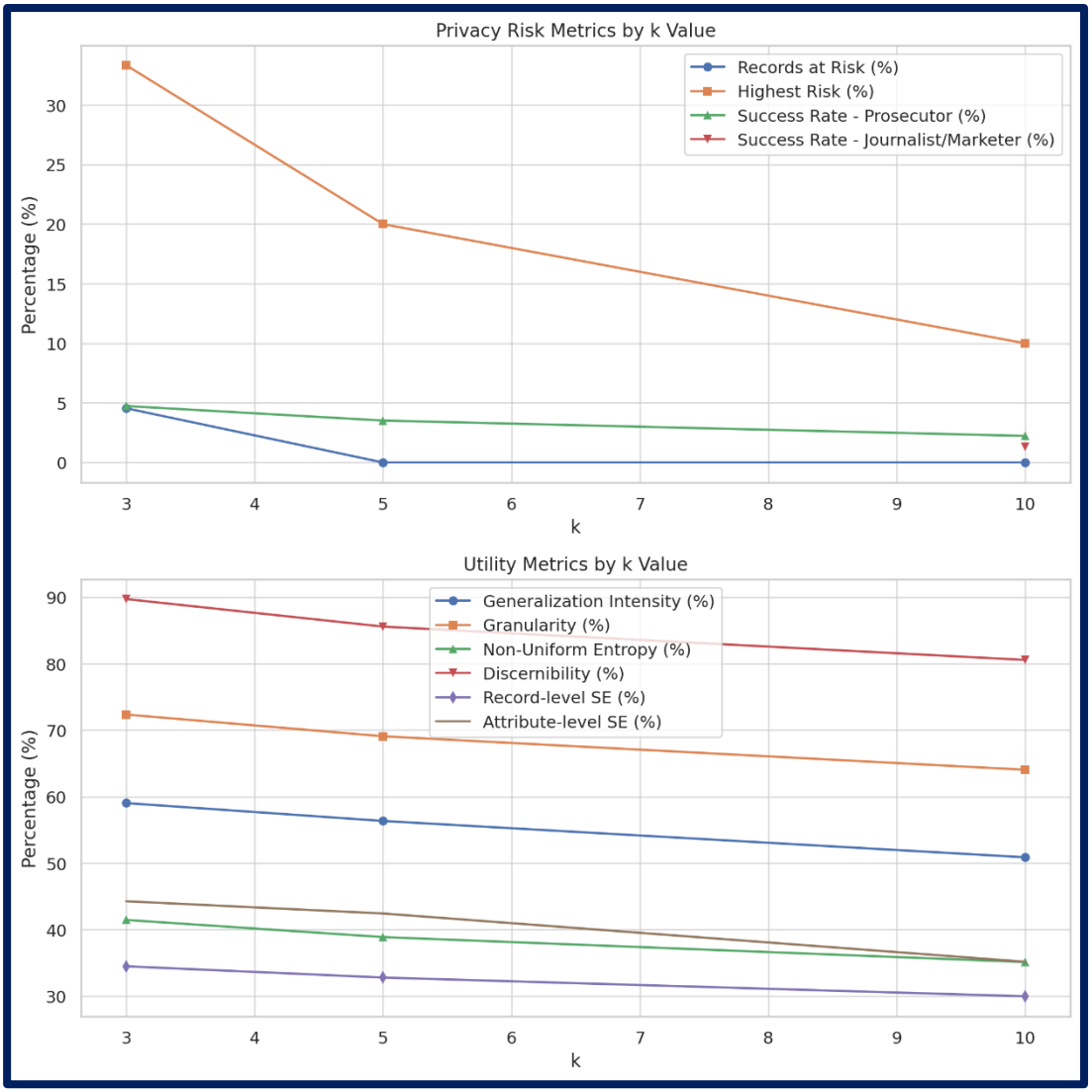

Figure 10 - Analysis of Privacy Risks and Data Utility Across 3 Different k-Anonymity Levels

4.6 Second Privacy Model Selected: l-Diversity

For this assignment, the second privacy model chosen was **ℓ-diversity**, a widely used extension of k-anonymity designed to provide stronger protection against **attribute disclosure**. While k-anonymity ensures that each record is indistinguishable from at least (k–1) others in terms of quasi-identifiers (QIDs), ℓ-diversity goes further by ensuring that the **sensitive attribute(s)** in each anonymized group also exhibit sufficient **diversity**.

ℓ-diversity addresses the limitations of k-anonymity in scenarios where equivalence classes may be homogeneous — that is, where all records in a group share the same sensitive value, making it possible to infer that value even when identity remains hidden.

In this model, all attributes were initially treated as QIDs, with the exception of salary, which was explicitly defined as a **sensitive attribute** due to its direct connection to personal financial information and potential for misuse. The presence of a sensitive attribute makes ℓ-diversity an appropriate and necessary model for ensuring privacy protection at the content level.

The ℓ-diversity model was selected for the following reasons:

- **Protection beyond identity disclosure:**
  While k-anonymity prevents identity recognition, it does not prevent **inference attacks** when sensitive values lack variation. ℓ-diversity guarantees that each anonymized group contains at least ℓ **well-represented sensitive values**, reducing the risk of sensitive information disclosure.
- **Compatibility with sensitive attributes:**
  Since salary is semantically sensitive and discretized into value ranges, applying ℓ-diversity ensures that groups are not only anonymized by identity but also by **value variability**, preventing homogeneity and background knowledge attacks.
- **Enhancing robustness of privacy guarantees:**
  By incorporating semantic diversity into the anonymization process, ℓ-diversity offers a **stronger and more realistic** privacy guarantee suitable for datasets where the sensitive attribute may be easily inferred.

Distinct ℓ-diversity was applied using a fixed value of **ℓ = 2.** This value was chosen to strike a balance between **privacy robustness** and **data utility**, as higher ℓ values typically increase generalization or suppression.

The anonymization was carried out using **ARX's Optimal search strategy**, which identifies the transformation that minimizes information loss while satisfying the ℓ-diversity constraint. The **Global Transformation** strategy was used to ensure uniformity across the dataset. This configuration promotes consistency in generalization and supports reliable evaluation of privacy risk and utility trade-offs.

4.7 Comparison of Re-identification Risk Before and After Anonymization Using Distinct l-Diversity

A comparative analysis between the **original dataset** (prior to anonymization) and the **distinct ℓ-diverse dataset** shows significant improvements in privacy protection:

Prosecutor Model

- **Before anonymization:**
  - Records at risk: **93.20%**
  - Highest risk: **100%**
  - Success rate: **80.34%**
- **After ℓ-diversity (ℓ = 2):**
  - Records at risk: **5.55%**
  - Highest risk: **50%**
  - Success rate: **3.47%**

Journalist Model

- **Before anonymization:**
  - Records at risk: **93.20%**
  - Highest risk: **100%**
  - Success rate: **80.34%**
- **After ℓ-diversity (ℓ = 2):**
  - Records at risk: **3.57%**
  - Highest risk: **50%**
  - Success rate: **2.28%**

Marketer Model

- **Before anonymization:**
  - Success rate: **80.34%**
- **After ℓ-diversity (ℓ = 2):**
  - Success rate: **2.28%**

These results clearly demonstrate that applying distinct ℓ-diversity significantly reduces re-identification risks across all attacker models. The **records at risk** dropped by over 87%, and **success rates** decreased from over 80% to just around **2–3%**, depending on the model. The highest possible re-identification risk was also **halved**, from 100% to 50%.
While not as strict as k-anonymity in eliminating records at risk (which dropped to 0% with k = 5), ℓ-diversity still provides **robust privacy guarantees**, particularly for protecting sensitive attributes from inference attacks.

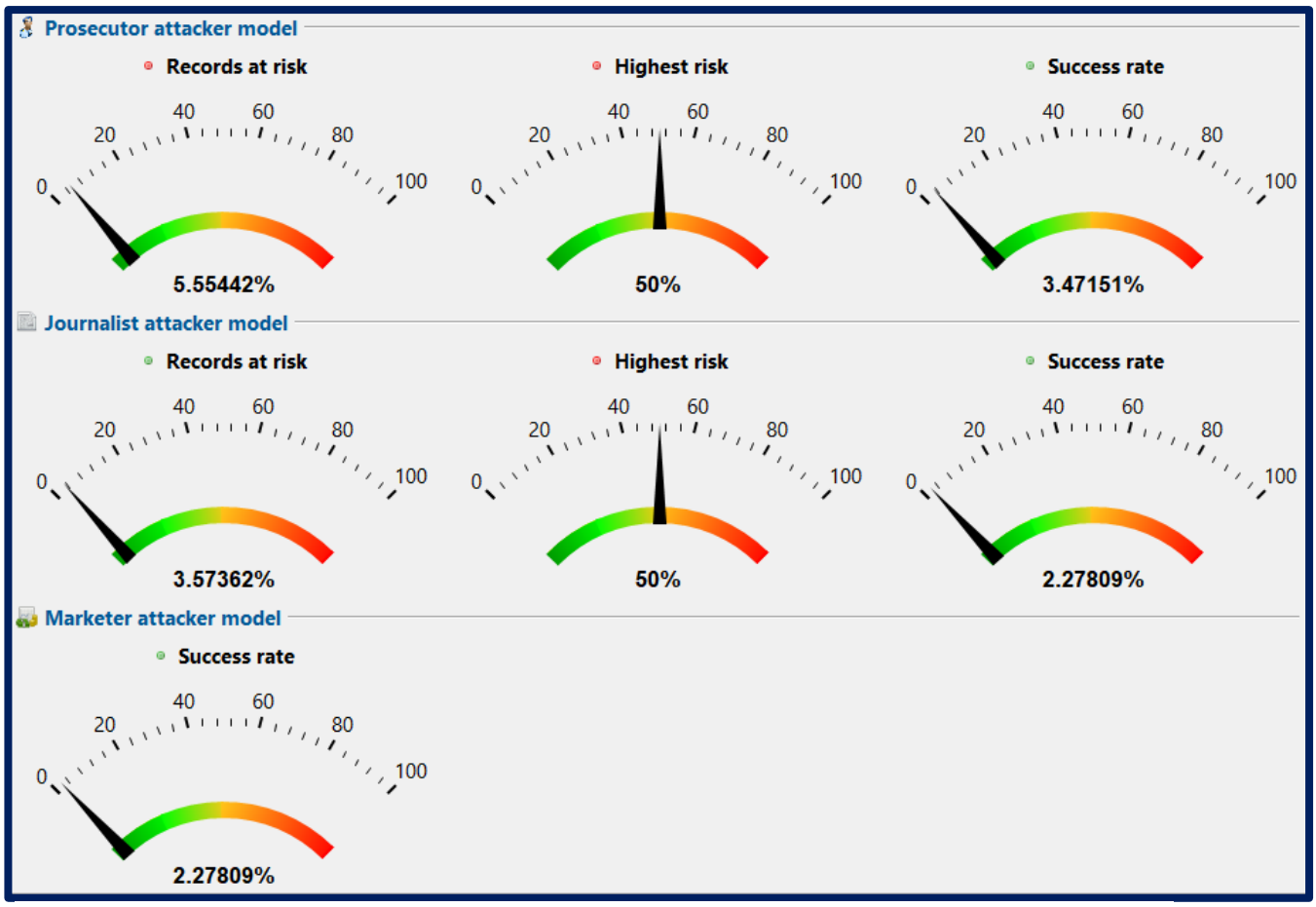

Figure 11 - Re-identification risk after applying distinct l-diversity with l = 2.

4.8 Additional risk metrics after applying distinct l-diversity

Metric	Value
Lowest prosecutor risk	0.19%
Records affected by lowest risk	10.93%
Average prosecutor risk	3.47%
Highest prosecutor risk	50%
Records affected by highest risk	1.18%
Estimated prosecutor risk	50%
Estimated journalist risk	50%
Estimated marketer risk	3.47%
Sample uniques	0%
Population uniques	0%
Population model	DANKAR

Table 5 - Additional Risk Metrics after anonymization with distinct l-diversity

A comparison between the original dataset and the anonymized version using **distinct ℓ-diversity (ℓ = 2)** reveals a significant improvement in privacy protection. In the original dataset, re-identification risk was alarmingly high: the **average prosecutor success rate** stood at **80.34%**, with **estimated risks** for both prosecutor and journalist attackers reaching **100%**. Additionally, **69.33% of records** were considered **unique** within the dataset, making them highly vulnerable to re-identification. The population uniqueness was also non-negligible at **4.17%**, indicating privacy risk even when the dataset is viewed as a sample of a broader population.

In contrast, after applying ℓ-diversity, the **average prosecutor risk** dropped sharply to **3.47%**, with **no records remaining unique** in either the sample or the estimated population. The lowest prosecutor risk improved from **5.56% to just 0.19%**, and the percentage of records affected by high risk (50%) fell from **69.33% to only 1.18%**. Furthermore, the **estimated success rates** for all attacker models were substantially reduced, dropping from **80–100%** to **just 3–50%**, depending on the model. These results confirm that ℓ-diversity effectively reduced both identity and attribute disclosure risks, transforming a highly vulnerable dataset into one that satisfies strong privacy guarantees.

4.10 Utility Evaluation – Distinct l-Diversity

Dataset-level quality

Model	Quality
Gen. intensity	56.45918%
Granularity	65.03913%
N.-U. entropy	35.97311%
Discernibility	83.55633%
Average class size	99.51723%
Record-level squared error	51.95168%
Attribute-level squared error	74.54975%
Aggregation-specific squared error	17.9032%

Figure 12 - Dataset-level quality

Following the application of distinct ℓ-diversity with ℓ = 2, the utility of the anonymized dataset was assessed using ARX's dataset-level quality models. The results reflect a well-balanced anonymization outcome, where the data retains analytical usefulness while providing protection against attribute disclosure.

The **generalization intensity** was **56.46%**, indicating that just over half of the attribute values were generalized during the anonymization process. This shows that a moderate level of abstraction was applied across the dataset to ensure that sensitive values appear with sufficient diversity within equivalence classes.

The **granularity** metric reached **65.04%**, suggesting that the data maintained a reasonable level of attribute specificity. Although some generalization occurred, a significant portion of the original value detail was preserved.

The **non-uniform entropy** was measured at **35.97%**, reflecting moderate information loss. This metric captures how well the anonymized dataset retains the original distribution of attribute values, and the result indicates that a substantial amount of useful variation was preserved despite anonymization.

The **discernibility score** was **83.56%**, showing that most records remain distinguishable within their respective anonymity groups. This is a positive indicator of utility, particularly for applications that rely on class-level distinctions.

The **average class size** was high at **99.51%**, which suggests that most records are grouped into equivalence classes of similar size, ensuring consistent treatment across the dataset.

Additional metrics like the **record-level squared error (51.95%)** and **attribute-level squared error (74.55%)** show a more noticeable degree of distortion, particularly at the attribute level, which is expected due to the more demanding constraints of ℓ-diversity.

In conclusion, the distinct ℓ-diversity model with ℓ = 2 effectively balances privacy protection and utility. While some increased generalization was necessary to satisfy the diversity requirement, the dataset retains a high degree of analytical value, making it suitable for exploratory and statistical analyses.

Attribute-level quality

Attribute	Data type	Missings	Gen. intensity	Granularity	N.-U. entropy	Squared error
sex	String	13.96697%	86.03303%	86.03303%	83.94465%	86.03303%
age	String	13.96697%	21.50826%	69.52164%	18.05433%	69.19554%
race	String	13.96697%	86.03303%	86.03303%	64.32523%	87.18075%
marital-status	String	13.96697%	43.01651%	57.14746%	49.07166%	3.35685%
education	String	100%	0%	0%	0%	0%
native-country	String	13.96697%	86.03303%	86.03303%	45.15463%	88.2663%
workclass	String	13.96697%	86.03303%	86.03303%	82.78373%	87.18902%
occupation	String	13.96697%	43.01651%	61.97767%	40.90396%	48.73565%

Figure 13 - Attribute-level quality

At the attribute level, ℓ-diversity (ℓ = 2) led to high generalization in attributes like **sex**, **race**, **native-country**, and **workclass** (over 86%), though **non-uniform entropy** remained high, preserving value diversity. Notably, **education** was fully suppressed, likely due to its uniqueness relative to the sensitive attribute. These results confirm that while privacy was strengthened, some attributes required substantial transformation, impacting their analytical utility.

4.11 Impact of Varying *l* in Distinct l-Diversity

In our case, the sensitive attribute **salary-class** has only **two possible values**:

- >50K
- <=50K

In **distinct ℓ-diversity**, each equivalence class must contain at least **ℓ distinct values** of the sensitive attribute.

Therefore:

- ℓ = 2 is the **maximum meaningful value** for this dataset.
- Trying ℓ = 3 (or more) would be **invalid**, because we can't have 3 distinct values in a domain with only 2.

Conclusion:

- ℓ = 2 is the **strongest possible** setting for **distinct ℓ-diversity** in our dataset.
- Testing with ℓ = 1 is valid but **not useful** — it would reduce the model to **basic k-anonymity**, offering **no protection against attribute disclosure**.
- So: ℓ = 2 is the **only non-trivial setting** we can meaningfully evaluate for this dataset under distinct ℓ-diversity.

4.12 Comparison between k-Anonymity (k=5) and Distinct ℓ-Diversity(l=2)

To conclude the evaluation, a comparison was made between the two privacy models applied: **k-anonymity (k = 5)** and **distinct ℓ-diversity (ℓ = 2)**. Although both models aim to reduce re-identification risk, they differ in their underlying goals and applicability.

Privacy Risk Mitigation

- **k-Anonymity (k = 5)** ensures that each individual is indistinguishable from at least four others based on quasi-identifiers. The resulting success rate dropped significantly to **3.52%**, with **0% of records at risk** under the Prosecutor and Journalist attacker models.
- **ℓ-Diversity (ℓ = 2)** extends this by ensuring **diversity in sensitive attribute values** within each equivalence class. While it maintained good privacy protection, the Prosecutor success rate was slightly higher at **3.47%**, and **5.55% of records** were still considered at risk. The highest risk also increased from **20% (k-anonymity)** to **50% (ℓ-diversity)**
.

This shows that while **k-anonymity offered stronger identity protection**, **ℓ-diversity adds protection against attribute disclosure**, particularly for the salary-class attribute.

Utility Comparison

In terms of utility:

- **k-Anonymity** achieved:
    - **Generalization intensity**: 56.35%
    - **Non-uniform entropy**: 38.88%
    - **Discernibility**: 85.59%
- **ℓ-Diversity** achieved:
    - **Generalization intensity**: 56.46%
    - **Non-uniform entropy**: 35.97%
    - **Discernibility**: 83.56%
    -

The utility values are relatively close, with k-anonymity offering slightly better entropy and discernibility, suggesting marginally better preservation of analytical patterns.

Applicability and Interpretation

- **k-Anonymity** is more generic and applicable when all attributes are quasi-identifiers, as in the first part of this assignment.
- **ℓ-Diversity**, however, becomes necessary when a **sensitive attribute is present** (in this case, salary-class) and its inference could lead to **attribute disclosure** — even if identity remains anonymous.

Conclusion

Both models effectively reduced re-identification risks, but with slightly different trade-offs:

- **k-Anonymity (k = 5)** offered **stronger identity protection** and marginally better utility.
- **ℓ-Diversity (ℓ = 2)** provided **additional protection** for the sensitive attribute, **at the cost of slightly higher risk metrics**.

Ultimately, the best model depends on the specific privacy requirements. In scenarios where **sensitive attributes are at stake**, **ℓ-diversity provides essential safeguards** beyond what k-anonymity alone can offer.

5. Parameter Sensitivity Analysis of the Privacy Model

In this section, we evaluate how varying specific parameters within the k-anonymity model and distinct l-diversity model affects the balance between privacy protection and data utility. Here the focus is on **internal anonymization parameters**, such as the **suppression limit**, **coding model**, and **attribute weights**. These parameters influence how generalization and suppression are applied and can significantly impact the outcome, even with *k* or l fixed.

To ensure consistency, all experiments in this section were performed with **k = 5 (for k-anonymity model)** and **l = 2 (for distinct l-diversity)**. For each configuration, re-identification risk and utility metrics were collected and analyzed using the ARX tool.

Plots	Purpose	Why it matters
Success Rate (%)	Privacy	Shows attacker success probability (intuitive: lower success rate → better privacy
Non-uniform Entropy (%)	Utility	Measures retained information (how much information is still present in the anonymized dataset)
Generalization Intensity (%)	Utility	Measures how much the dataset was altered (higher % → more distortion)

Table 6 - Metrics used to understand the impact of privacy model parameters

5.1 Suppression Limit

In this section, we investigate how varying the **suppression limit** affects the trade-off between **privacy protection** and **data utility**. The suppression limit defines the maximum percentage of records that can be suppressed to achieve the desired privacy level.

To assess this impact, we analyzed the dataset under multiple suppression thresholds: **2.5%, 5%, 10%, 20%, 50%, and 100%**. For each configuration, we evaluated privacy risk using the **success rate** of attacker models (Prosecutor, Journalist, and Marketer), and utility using **Non-uniform Entropy** and **Generalization Intensity**.

The results are presented in the form of line plots, which show how these metrics evolve as the suppression limit increases. This allows us to identify an effective suppression threshold that minimizes privacy risk while retaining acceptable levels of data quality.

5.2 k-Anonymity (k = 5) – Suppression Limit

Suppression Limit (%)	Prosecutor Success Rate (%)	Journalist & Marketer Success Rate (%)	Gen. Intensity (%)	N.-U. Entropy (%)
2.5	1.19943	0.76462	46.34	28.03
5	1.45061	1.45061	53.15	29.62
10	2.19365	2.19365	55.05	34.22

Suppression Limit (%)	Prosecutor Success Rate (%)	Journalist & Marketer Success Rate (%)	Gen. Intensity (%)	N.-U. Entropy (%)
20	3.51586	3.51586	56.35	38.88
50	3.51586	3.51586	56.35	38.88
100	3.51586	3.51586	56.35	38.88

Table 7 - Suppression Limit and its Metrics

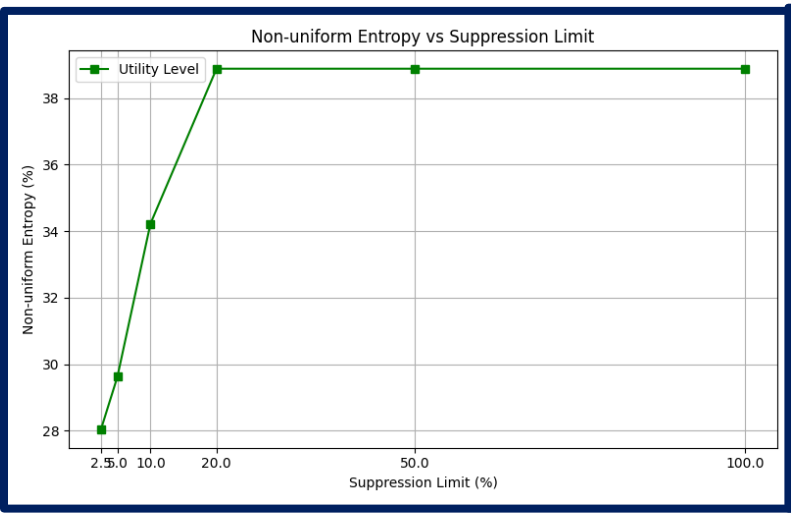

Figure 14 - Plot 1: Non-uniform Entropy vs Suppression Limit

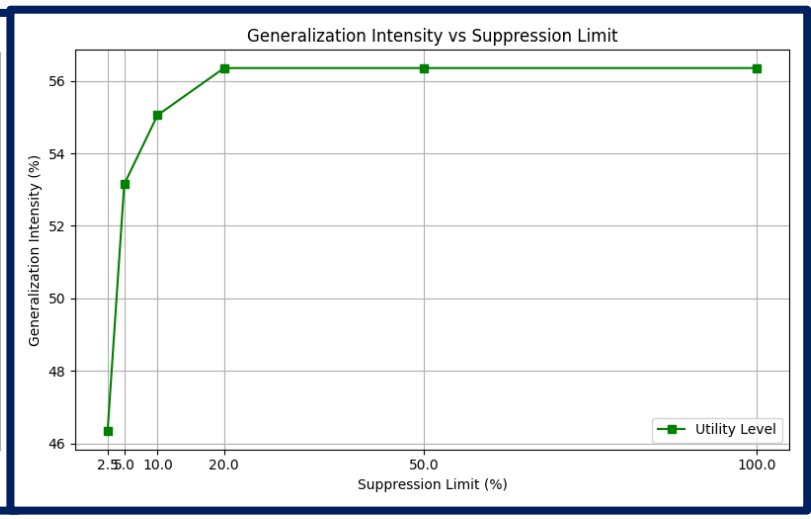

Figure 15 - Plot 2: Generalization Intensity vs Suppression Limit

Figure 14 – *Non-uniform Entropy as a function of the suppression limit applied under the k-anonymity model (k = 5).*

This plot illustrates how the **utility** of the dataset, measured via **non-uniform entropy**, varies with increasing suppression limits (2.5% to 100%). A higher entropy score indicates greater retention of information. As the suppression limit increases:

- Entropy improves steadily from 28% to ~39%.
- The metric plateaus around **20%**, indicating that allowing further suppression yields diminishing gains in utility.

X-axis: Suppression Limit (%)
Y-axis: Non-uniform Entropy (%)
Metric Type: Utility

Figure 15 – *Generalization intensity measured at different suppression limits under the k-anonymity model (k = 5).*

This metric indicates how generalized the quasi-identifiers are — higher values mean more abstraction and less data precision. As suppression increases:

- Generalization intensity rises from 46% to 56%.
- Like entropy, this too stabilizes around the 20% threshold, confirming a balance between data utility and anonymization effort.

X-axis: Suppression Limit (%)
Y-axis: Generalization Intensity (%)
Metric Type: Utility

Generalization intensity and **non-uniform entropy** increased **slightly** with suppression, indicating **more aggressive generalization** at lower suppression limits.

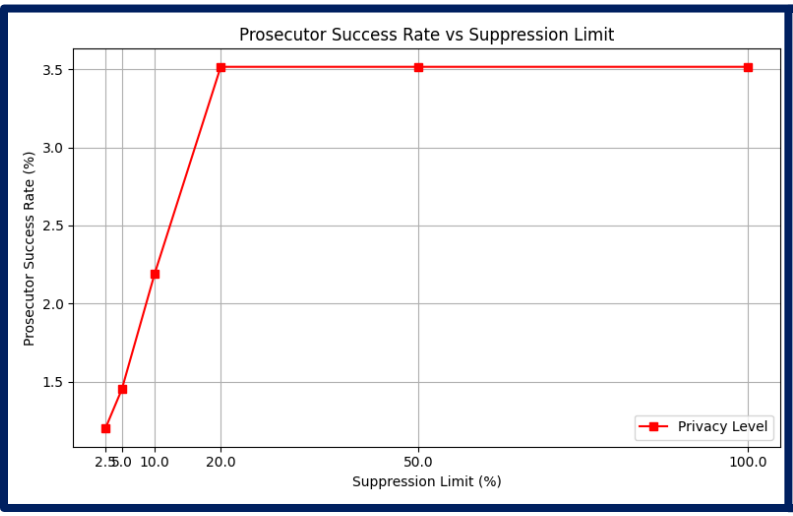

Figure 16 - Plot 3: Prosecutor Success Rate vs Suppression Limit

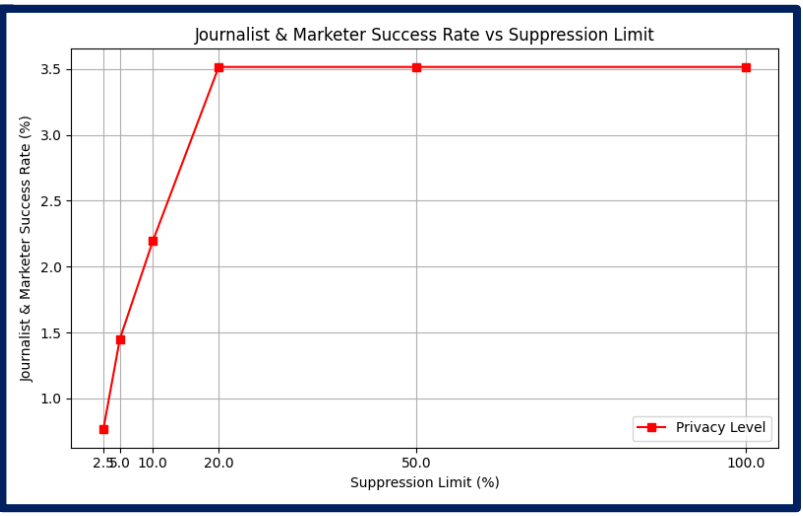

Figure 17 - Plot 4: Journalist & Marketer Success Rate vs Suppression Limit

Figure 16 – *Prosecutor attacker model success rate for increasing suppression limits (k = 5).*

This plot shows how the success rate of a **targeted re-identification attack** drops as more suppression is allowed:

- At 2.5% suppression, the success rate is **1.20%**
- It gradually increases to **3.51%** at 20% and **remains stable** beyond that
- This plateau suggests that further increasing the suppression limit **does not reduce privacy risk** any further under this model

X-axis: Suppression Limit (%)
Y-axis: Prosecutor Success Rate (%)
Interpretation: Lower success rate means stronger privacy guarantees

Figure 17 – *Success rate for the Journalist and Marketer attacker models under varying suppression limits (k=5).*

These two models reflect more realistic or mass-scale attacks (e.g., probabilistic matching or uncertain presence in the dataset). In all tested suppression settings:

- The Journalist and Marketer models produced **identical success rates**
- Their values also **plateaued at 3.51%** after 20% suppression
- The largest privacy gain occurs between 2.5% and 20%, confirming the effectiveness of minimal suppression when properly tuned

X-axis: Suppression Limit (%)
Y-axis: Success Rate (%)
Interpretation: Effective suppression limits **below 20%** can reduce re-identification risk drastically

5.3 Distinct l-Diversity (l = 2) – Suppression Limit

Suppression Limit (%)	Prosecutor Success Rate (%)	Journalist & Marketer Success Rate (%)	Gen. Intensity (%)	N.-U. Entropy (%)
2.5	0.83	0.53	39.66	20.28
5	1.19	1.19	48.06	26.28
10	2.52	2.52	52.51	33.24
20	3.47	2.28	56.46	35.97
50	3.47	2.28	56.46	35.97
100	3.47	2.28	56.46	35.97

Table 8 - Suppression Limit and its Metrics

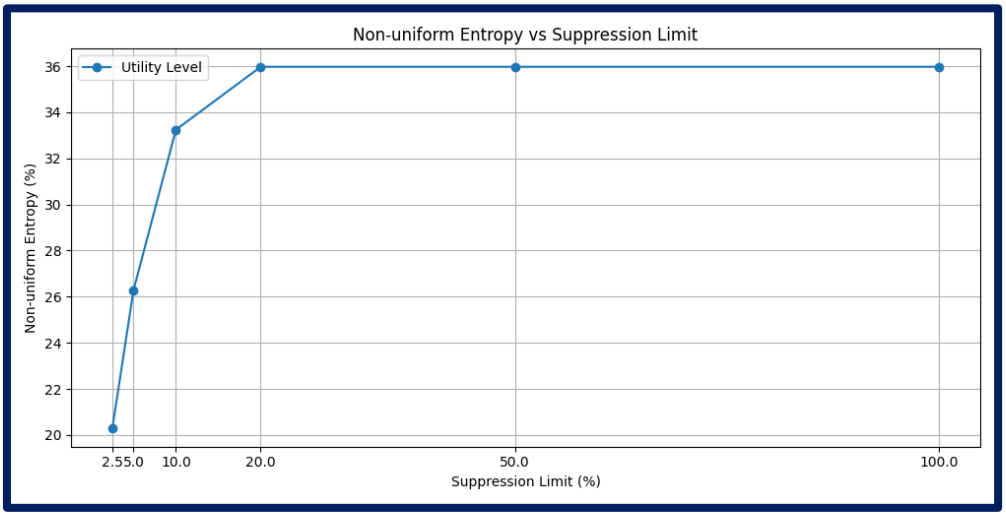

Figure 18 - Plot 5: Non-uniform Entropy vs Suppression Limit

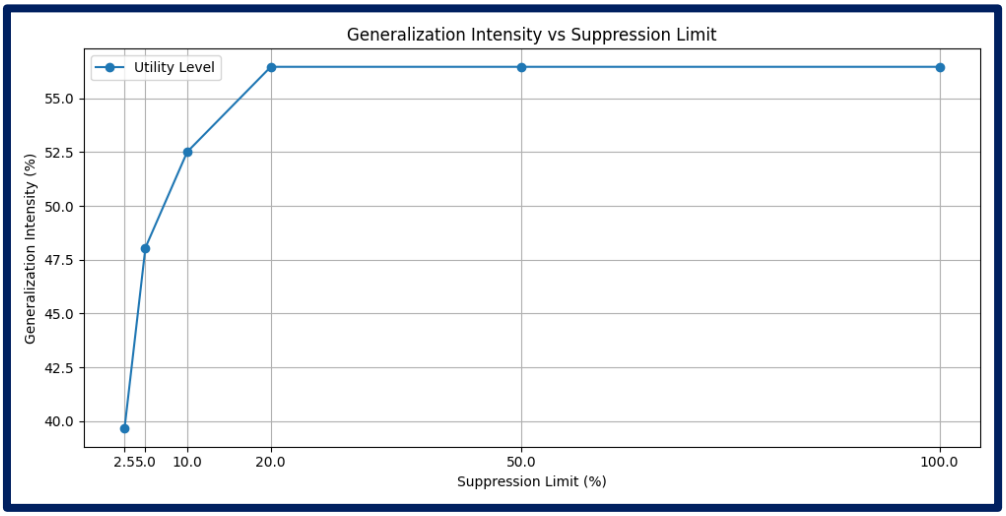

Figure 19 - Plot 6: Generalization Intensity vs Suppression Limit

Figure 18 – *Non-uniform Entropy as a function of the suppression limit applied under the distinct l-diversity model (l = 2).*

Non-uniform entropy measures the retention of information diversity within quasi-identifiers. A higher value suggests better preservation of the dataset's internal distribution:

- At **2.5% suppression**, the N.-U. entropy is about **20.28%,** which is quite low due to strict constraints.
- As suppression is relaxed to **10%**, entropy rises sharply to **33.24%,** and reaches **35.97%** at **20%**, where it then stabilizes.
- Utility measured by entropy also plateaus after the 20% threshold.

This suggests that a suppression limit of **20%** provides a strong balance between data utility and privacy, maximizing entropy without further gains afterward.

X-axis: Suppression Limit (%)
Y-axis: Non-uniform Entropy (%)

Figure 19 - *Generalization intensity measured at different suppression limits under the distinct l-diversity (l=2).*

The first plot shows how Generalization Intensity changes with increasing suppression limits. Generalization intensity reflects the extent to which data values are generalized to satisfy the privacy constraint:

- At a low suppression limit of 2.5%, generalization intensity is approximately **39.66%,** meaning fewer generalizations are permitted, requiring stricter suppression.
- As the suppression limit increases to 10% and then 20%, generalization intensity steadily increases, peaking at **56.35%** from 20% onward.
- From **20% to 100%,** the generalization intensity plateaus, indicating that ARX no longer needs to generalize more once a certain suppression flexibility is allowed.

This indicates that beyond a suppression limit of 20%, additional suppression does not lead to further generalization gains, suggesting diminishing returns in utility trade-offs.

X-axis: Suppression Limit (%)
Y-axis: Generalization Intensity (%)

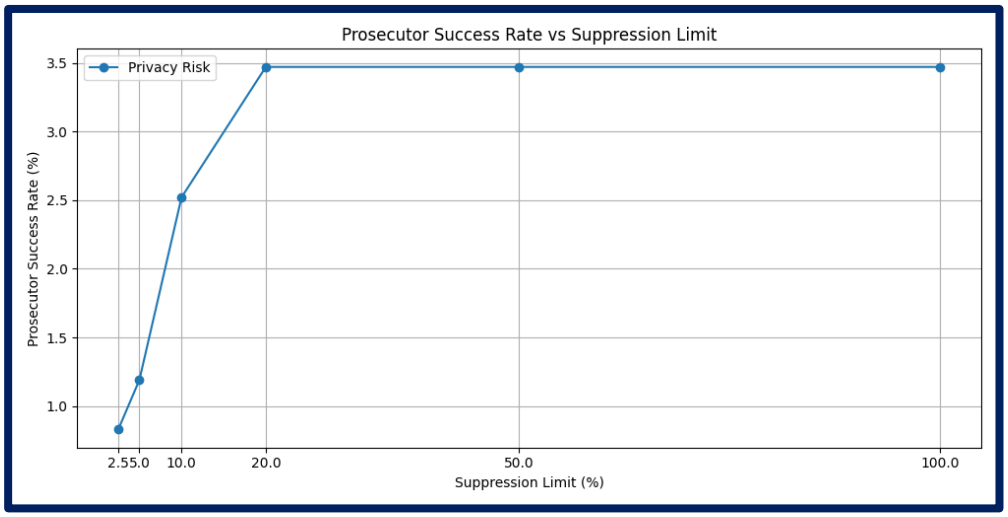

Figure 20 - Plot 7: Prosecutor Success Rate vs Suppression Limit

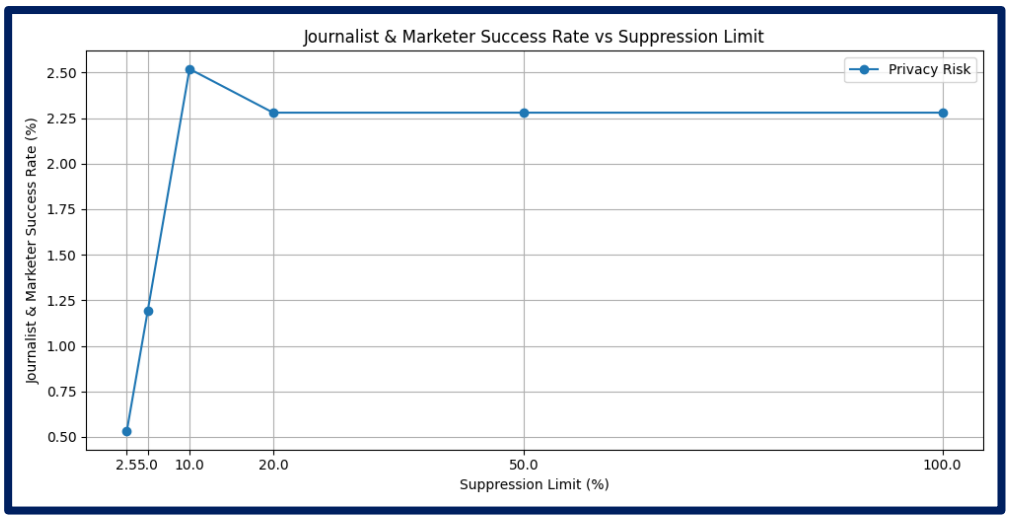

Figure 21 - Plot 8: Journalist & Marketer Success Rate vs Suppression Limit

Figure 20 – *Prosecutor attacker model success rate for increasing suppression limits (l = 2).*

- This risk curve rises more steeply than the marketer/journalist model.
- Starts at 0.83% (2.5% suppression) and increases to 3.52% by 20% suppression, then plateaus.
- It indicates that for attackers who know the target is in the dataset, suppression limit has a clearer effect on risk control.

- **X-axis:** Suppression Limit (%)
- **Y-axis:** Success Rate (%)

Figure 21 – *Success rate for the Journalist and Marketer attacker models under varying suppression limits (l=2).*

- Risk increases from very low values (~0.53%) at 2.5% suppression to a peak of 2.52% at 10%, then stabilizes.
- Despite the increase, the risk values remain low, showing that the model maintains strong protection even at lower suppression limits.

- **X-axis:** Suppression Limit (%)
- **Y-axis:** Success Rate (%)

5.4 - Effect of Coding Model on Privacy Risk and Utility

In this section, we examine the impact of the **coding model** on the **privacy risk** of the dataset. The coding model determines how generalization is applied to quasi-identifiers:

- **Global recoding** applies the same generalization level to all values of an attribute across the entire dataset.
- **Local recoding** allows context-specific generalization, adapting to subsets of the data and enabling more flexible transformations.

In our analysis, we used the ARX tool to apply both **k-anonymity (with k = 5)** and **distinct l-diversity (with l = 2)**. The **Global Recoding** was implemented using the default **"Global Transformation"** option, while the **Local Recoding** was applied using the **"Local Transformation with iterations = 10"**. For each configuration, privacy risk was assessed via the **success rate** of attacker models, and utility was measured through **Non-uniform Entropy**, **Generalization Intensity**, and **Discernibility**.

5.5 k-Anonymity (k = 5) - Effect of Coding Model on Privacy Risk and Utility

Metric	Global Transformation	Local Transformation with iterations=10
Success Rate	3.52%	12.42%
Generalization Intensity	56.35%	80.26%
Non-uniform Entropy	38.88%	58.30%
Discernibility	85.59%	99.84%

Table 9 - Metrics (Global vs Local Transformation)

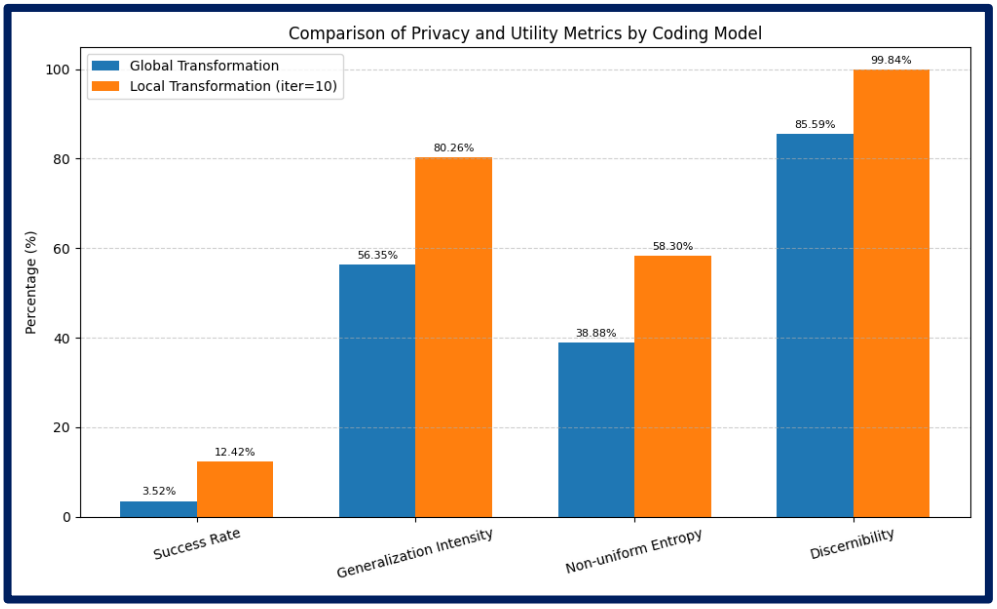

Figure 22 - Side-by-side bar chart comparing success rate, entropy, generalization and discernibility for global vs local coding (k-anonymity)

The results clearly illustrate the trade-off between privacy and utility that arises from the choice of coding model:

- Under **global recoding**, the re-identification success rate was significantly lower (3.52%), reflecting stronger privacy protection. However, this came at the cost of moderate utility loss due to uniform generalization applied across all records.
- In contrast, **local recoding** preserved more specific data, as indicated by higher utility metrics: generalization intensity increased to 80.26%, non-uniform entropy rose to 58.30%, and discernibility reached 99.84%. This gain in utility,

however, resulted in a substantially higher re-identification success rate (12.42%), thus reducing overall privacy protection.

These findings underscore the need to carefully balance privacy risk mitigation with analytical utility when selecting a coding model for data anonymization.

5.6 Distinct l-Diversity (l = 2) - Effect of Coding Model on Privacy Risk and Utility

Metric	Global Transformation	Local Transformation (iter=10)
Success Rate	3.47%	16.81%
Generalization Intensity	56.46%	74.21%
Non-uniform Entropy	35.97%	56.15%
Discernibility	83.56%	98.38%

Table 10 - Metrics (Global vs Local Transformation)

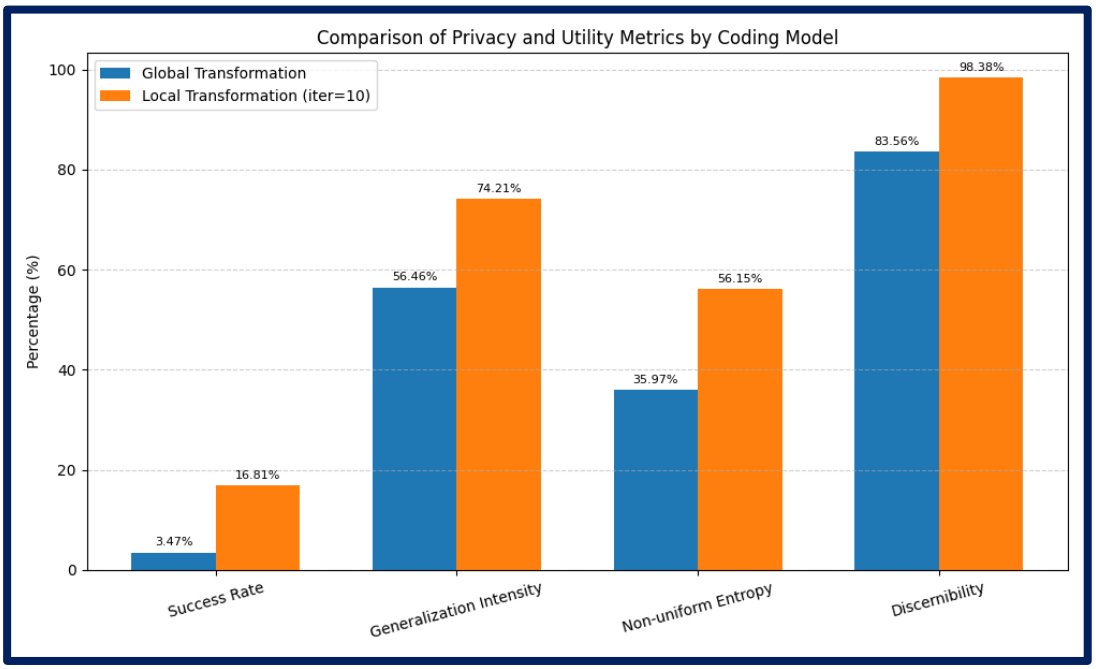

Figure 23 - Side-by-side bar chart comparing success rate, entropy, generalization and discernibility for global vs local coding (distinct l-diversity)

The results of the comparison between **Global Transformation** and **Local Transformation (iterations = 10)** under the **distinct 2-diversity model** highlight a clear trade-off between privacy and utility.

The **success rate** (proxy for re-identification risk) increased significantly from **3.47%** under global transformation to **16.81%** with local transformation. This indicates that the localized flexibility in generalization allowed more precise preservation of data, but at the cost of reduced privacy protection.

In terms of utility, **Local Transformation** outperformed the global approach across all measured metrics. **Generalization intensity** increased from **56.46%** to **74.21%**, and **non-uniform entropy** rose from **35.97%** to **56.15%**, reflecting better preservation of data variability and distribution. Similarly, **discernibility** improved from **83.56%** to **98.38%**, showing that records remained more individually distinguishable within the dataset.

Overall, local transformation delivers higher utility but introduces greater privacy risks, making it suitable in scenarios where data usefulness is prioritized and slightly elevated risk is acceptable. Conversely, global transformation remains the better choice for use cases demanding stronger anonymity guarantees.

5.7 Attribute Weights – Customizing the Importance of Quasi-Identifiers

In ARX, ***attribute weights*** allow fine-grained control over how each quasi-identifier (QID) is treated during anonymization. Rather than assuming all attributes have equal importance, this feature enables the user to specify which attributes are **more critical to preserve** and which can be **more flexibly generalized**.

Objective:

The main goal of using attribute weights is to achieve a **better balance between privacy and utility**, based on the analytical needs of the dataset. This is especially useful when some attributes carry more relevance for downstream analysis, while others can tolerate greater abstraction.

How It Works:

When searching for an optimal transformation that satisfies a privacy model, ARX uses the assigned weights to guide the generalization process:

- Attributes with **higher weights** are prioritized for preservation, resulting in **less generalization**.
- Attributes with **lower weights** are **more likely to be generalized**, helping to meet the privacy constraint with **minimal impact on more important attributes.**

Why Use Attribute Weights?

Adjusting attribute weights is particularly helpful when:

- Certain attributes are **crucial for analysis** and must retain detail.
- Other attributes are **less informative** or more **easily generalizable**.
- The user wants to **preserve data utility selectively**, rather than applying uniform generalization across all fields.

By tuning attribute weights, the anonymization process becomes more aligned with **the analytical goals and data context**, improving the **overall usefulness** of the anonymized dataset while still respecting privacy constraints.

5.8 k-Anonymity (k = 5) – Attribute Weights

By default, ARX treats all quasi-identifiers as equally important during the anonymization process. However, in many practical scenarios, certain attributes are more valuable for downstream analysis, while others are less critical or more redundant.

Quasi-identifiers were grouped into three importance levels, and weights were assigned accordingly:

Attribute	Assigned Weight	Justification
age	1.0	Crucial for demographic analysis and strongly correlated with many target variables, including salary.
education	1.0	A strong predictor of income and essential for socioeconomic modeling.
native-country	1.0	Relevant in studies involving migration, inequality, or regional income trends.
marital-status	0.75	Informative but often overlaps with age or education; assigned moderate importance.
race	0.75	May be relevant in fairness or demographic studies but has low cardinality and can be generalized without major utility loss.
sex	0.75	Low variability (binary), somewhat redundant with other socio-demographic attributes; kept at moderate importance for potential gender-based analysis.

Attribute	Assigned Weight	Justification
workclass	0.25	High cardinality and often redundant with occupation and education; suitable for generalization to preserve other attributes.
occupation	0.25	Granular and correlated with both education and salary-class; generalization has low analytical cost.
salary-class	0.25	Though sensitive in other models, here it was treated as a QID. Already aggregated (<=50K, >50K), it tolerates generalization without significant utility loss.

Table 11 - Attribute Weights assigned and justification

These weights were applied using the ARX interface under the "Attribute Weights" tab before executing anonymization under the **k-anonymity model with k = 5**. The optimizer then used these weights to prioritize which attributes to preserve more precisely while still achieving the privacy constraint.

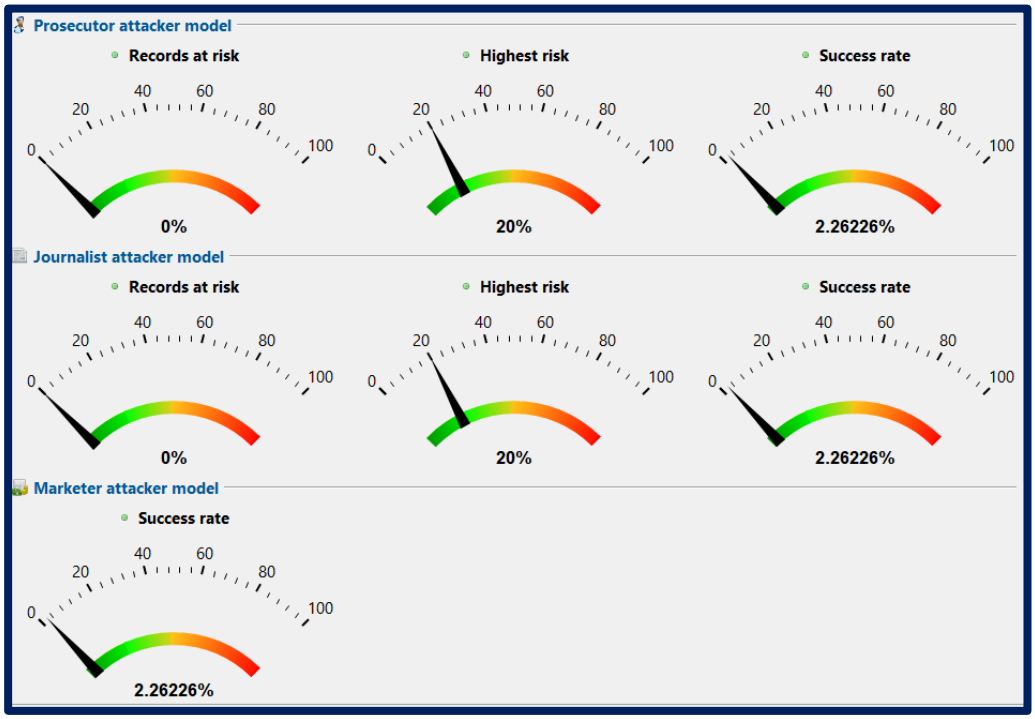

Figure 24 - Re-identification risk after applying attributes weights

Dataset-level quality

Model	Quality
Gen. intensity	56.87807%
Granularity	61.12068%
N.-U. entropy	32.53491%
Discernibility	83.69279%
Average class size	99.24511%
Record-level squared error	52.02879%
Attribute-level squared error	81.20482%
Aggregation-specific squared error	1.29079%

Figure 25 - Attribute-level utility after applying attribute weights

Metric	Percentage
Success Rate	2.26%
Records at Risk (all)	0%
Highest Risk	20%

Table 12 - Metrics after applying attribute weights

Metric	Value	Interpretation
Generalization Intensity	56.88%	Similar to the original (56.35%), showing stable overall generalization effort.
Non-uniform Entropy	32.53%	Slight drop (vs. 38.88%) due to increased suppression of less weighted attributes.
Discernibility	83.69%	Minor change, indicating most records remain distinguishable.
Attribute-level Squared Error	81.20%	Higher value reflects targeted utility preservation on key attributes.

Table 13 - Metrics after applying attribute weights

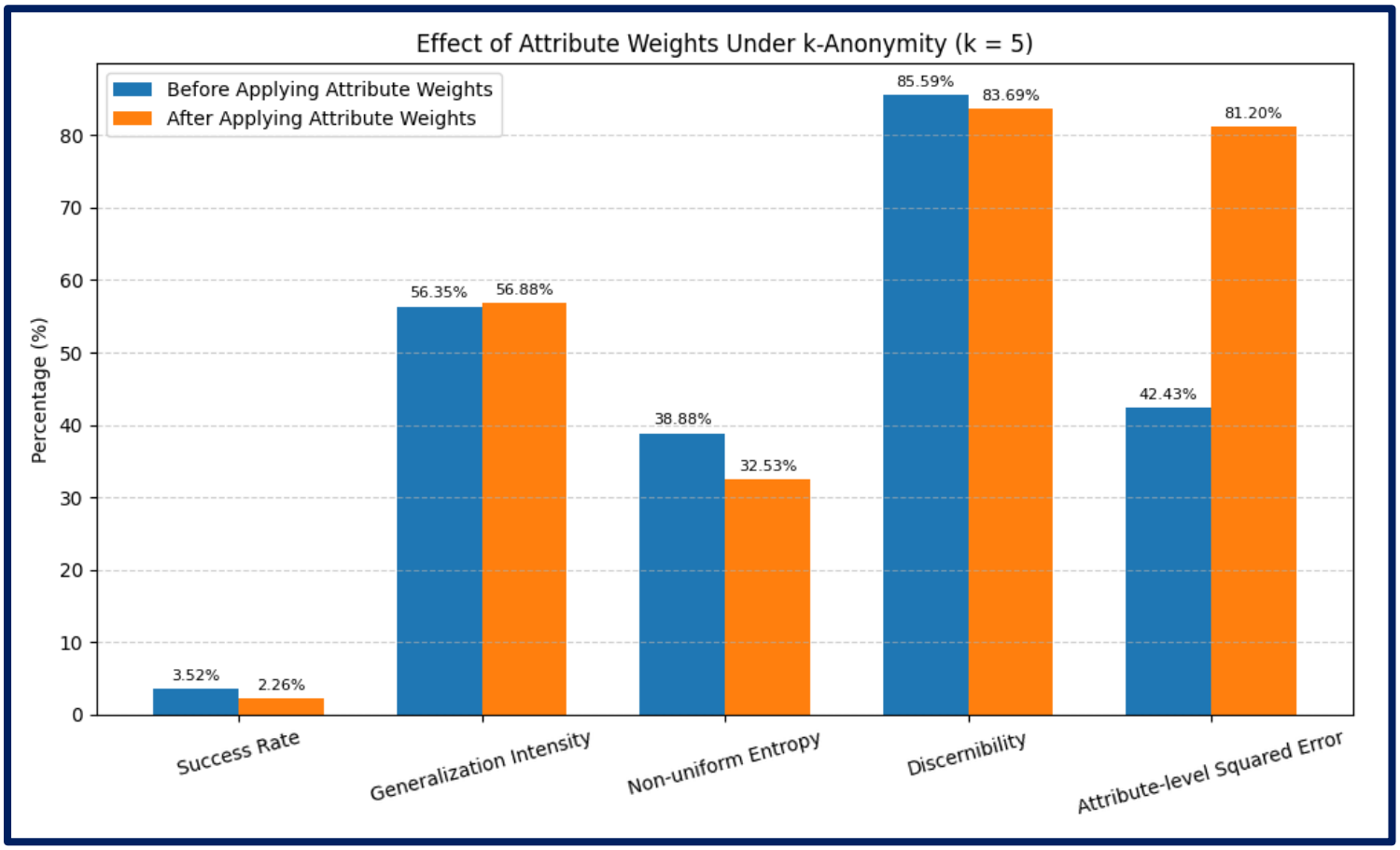

Figure 26 - Impact of Attribute Weights on Privacy Risk and Utility

Applying attribute weights in the context of k-anonymity (k = 5) led to a measurable improvement in privacy without compromising overall data utility. The success rate of re-identification attacks dropped from **3.52% to 2.26%**, indicating enhanced privacy protection. This reduction, though modest, is particularly valuable when considering the context of sensitive applications such as income prediction.

From a utility perspective, the **Generalization Intensity** remained virtually unchanged (**56.35% → 56.88%**), demonstrating that privacy gains did not come at the cost of significantly broader generalization. Interestingly, **Non-uniform Entropy** experienced a slight decline (**38.88% → 32.53%**), suggesting a reduction in variability across attribute distributions — a

predictable consequence of assigning lower weights to less informative features.

On the other hand, **Attribute-level Squared Error** increased substantially (**42.43% → 81.20%**), which may seem counterintuitive at first. However, this reflects the tool's shift in prioritization: by allowing greater distortion in less important attributes (e.g., race, sex), the algorithm preserved the integrity of more critical attributes like **salary-class** and **education**, which are essential for the dataset's analytical purpose.

Discernibility slightly decreased (**85.59% → 83.69%**), still maintaining a high level of distinguishability between records.

The application of attribute weights successfully balanced the privacy-utility trade-off, preserving meaningful structure where it mattered most and allowing harmless distortion elsewhere. This highlights the strategic advantage of incorporating domain knowledge into the anonymization process. When used thoughtfully, attribute weighting proves to be a powerful tool for tailoring anonymization to specific analytical goals.

5.9 – Distinct l-Diversity (l = 2) – Attribute Weights

In this section, we applied attribute weights under the **distinct l-diversity model** (with *l = 2*) to prioritize utility on the most relevant quasi-identifiers while protecting sensitive information.

The weights assigned to quasi-identifiers were the same as in the previous k-anonymity configuration. This consistency allows a fair comparison between both models regarding how attribute weighting impacts privacy-utility trade-offs. However, the key difference is that **salary-class is now treated as a sensitive attribute** under the l-diversity model. As a result, it is no longer included in the set of quasi-identifiers, and no weight was assigned to it. Instead, l-diversity ensures that within each equivalence class, there is sufficient diversity of salary-class values to protect against inference attacks.

Metric	Before Applying Attribute Weights	After Applying Attribute Weights
Success Rate	3.47 %	2.62%
Records at risk	5.55%	2.35%
Generalization Intensity	56.46%	55.90%
Non-uniform Entropy	35.97	32.36%
Discernibility	83.86%	82.41
Attribute-level Squared Error	74.55%	82.06%

Table 14 - Impact of Attribute Weights on Privacy Risk and Utility (2-Diversity)

After applying attribute weights under the distinct l-diversity model (l = 2), we observed a moderate improvement in privacy with relatively stable utility levels.

The **success rate** of all attacker models dropped from **3.47% to 2.62%**, indicating a reduced risk of successful re-identification. This suggests that emphasizing protection for more sensitive or analytically critical attributes—like salary-class—can lead to meaningful privacy gains.

Utility metrics also showed slight variations. The **Generalization Intensity** remained nearly constant (from **56.46% to 55.90%**), meaning that the overall level of generalization applied to the dataset did not increase substantially. However, **Non-uniform Entropy** decreased modestly (from **35.97% to 32.36%**), reflecting a slight reduction in distributional richness, which is a common trade-off when prioritizing certain attributes for privacy.

Discernibility, which measures how distinguishable records remain post-anonymization, dropped marginally from **83.56% to 82.41%**, indicating minimal impact on overall dataset granularity. On the other hand, the **Attribute-level Squared Error** increased from **74.55% to 82.06%**, suggesting that higher distortion occurred on specific attributes—precisely those we intentionally prioritized with higher weights.

In summary, applying attribute weights under distinct l-diversity helped reduce privacy risk without drastically

compromising overall utility. These results validate the effectiveness of attribute weighting when domain knowledge is available to guide which variables should be protected more aggressively.

6. Conclusion

After applying and evaluating two privacy models — **k-anonymity** and **ℓ-diversity** — we concluded that **k-anonymity with k = 5** provided the **best overall results in terms of balancing privacy protection and data utility**. This decision is supported by the following points:

- **Privacy**: With k = 5, the re-identification success rate dropped to **3.52%**, and **0% of records remained at crisk**, outperforming the ℓ-diversity model, where 5.55% of records were still vulnerable.
- **Utility**: The k = 5 model maintained **higher entropy (38.88%)** and **discernibility (85.59%)**, preserving the analytical value of the dataset more effectively.
- The choice of **k = 5** — instead of 3 or 10 — was based on the trade-off observed:
    - k = 3 did not offer sufficient protection (risk up to 33.33%).
    - k = 10 caused excessive generalization, leading to reduced utility without significantly improving privacy.
    - Therefore, **k = 5 offered the best compromise** between minimizing disclosure risk and preserving data utility.

In addition, we evaluated the effect of different **suppression limits**. A **20% suppression threshold** was identified as optimal: it provided a clear improvement in utility metrics (entropy and granularity), while keeping the re-identification risk low. Beyond this point, higher suppression limits did not yield meaningful gains, making 20% the most efficient choice (for both models).

We also compared two **coding models**: **Global Transformation** and **Local Transformation**. Although local transformation offered slightly higher utility, it also resulted in a much higher re-identification success rate. As our priority is to ensure robust privacy protection, we selected Global **Transformation**, which maintained a lower success rate and thus stronger anonymization guarantees (for both models)

Finally, we explored the use of **attribute weights** to prioritize the protection of semantically important variables, particularly those relevant for downstream analysis. After applying attribute weights, we observed an improvement in privacy-utility trade-offs:

- Under k-anonymity, the **success rate dropped** (from **3.52% to 2.26%**) and **utility increased**, confirming that the weighted configuration outperformed the unweighted one.
- Under ℓ-diversity, the **success rate also improved** (from **3.47% to 2.62%**) with attribute weights, although not as significantly as with k-anonymity.
- Comparing the weighted versions of both models, **k-anonymity with attribute weights** continued to outperform ℓ-diversity, confirming its robustness even after customization.

In conclusion, the submitted ARX project corresponds to the configuration using:

- **k-anonymity with k = 5**,
- **20% suppression limit**,
- **Global Transformation**,

- **Attribute weights** as selected in Section 5.8.

This configuration demonstrated the **best overall trade-off** between privacy protection and data utility for this dataset.